The Magnificent Conundrum

Abstract—**In recent years, MAG7 has dominated equity market capitalization and returns, challenging the assumptions of traditional financial theory and practice. This paper leverages techniques including β decomposition, DCC-GARCH, and CCC-EGARCH models to examine the impact of market concentration on the stock and options markets.**

Our stock market analysis shows that MAG7 represents a distinct risk factor, and a separated β model that accounts for this has better explanatory and forecasting power. In the options market, our findings indicate that concentration risk remains unpriced. However, stress tests in both markets underscore the need for improved risk management as concentration risk increases.

Finally, we evaluate trading strategies in response to these insights and find that a long-short, industry-neutral, MAG7-β-neutral strategy delivers the best performance, aligning with our analysis.

I. Introduction

The growing concentration of a select group of stocks, commonly referred to as the "Magnificent 7" (Apple, Microsoft, Alphabet, Amazon, Nvidia, Meta, and Tesla), within major equity indices such as the S&P 500 and NASDAQ 100 has significant implications for both financial theory and investment practice. For brevity, we denote this group as **MAG7** for the remainder of the paper. This paper examines how this increase in market concentration challenges traditional assumptions regarding market β, option pricing, and long-short equity strategies. Specifically, we analyze these effects through three key perspectives: (1) the decomposition of β and its implications for systematic risk, (2) the impact of concentration on index option pricing, including premium adjustments, implied correlation dynamics, and market crash risk, and (3) modifications to traditional long small-cap / short large-cap strategies in response to heightened concentration risk. Our methodology integrates empirical analysis and advanced time-series modeling to assess these evolving market dynamics. Our findings underscore the need for a paradigm shift in financial modeling, particularly in three areas: β decomposition, to distinguish the effects of dominant mega-cap stocks from broader market movements; index option pricing adjustments, to better capture implied correlation risk; and factor-based long-short strategies, which require structural refinements to mitigate concentration risk and appropriately account for mega-cap dominance.

II. Literature Review

Dennis, Mayhew, and Stivers (2006) found that the relationship between stock returns and systematic volatility innovations is significantly negative, whereas the relationship with idiosyncratic volatility innovations is near zero. This suggests that asymmetric volatility is mainly driven by systematic market-wide factors rather than firm-level effects and that innovations in implied volatility serve as reliable proxies for expected stock volatility changes. Driessen, Maenhout, and Vilkov (2009) studied the pricing of correlation risk in the stock options market. They found that a rise in overall correlation reduces investors' benefits from diversification and increases market risk. Index option prices include an additional premium for increasing correlations, making them more expensive than individual stock options. Buss and Vilkov (2012) concluded that option-implied correlations can be used to construct forward-looking market βs, which exhibit a strong and monotonic risk-return relationship compared to traditional rolling-window βs. Driessen, Maenhout, and Vilkov (2013) found that the implied correlation calculated from index options is generally higher than the realized correlation, indicating the presence of a correlation risk premium in index options. Further, implied correlation has significant predictive power for future stock market returns. Decker and Ferla (2023) examined the use of the DCC-GARCH model in portfolio optimization. They incorporated shrinkage methods and composite likelihood estimation to address large-scale portfolio allocation issues. While DCC-GARCH models adapted well to changes in the correlation structure, they failed to create a consistently better trading strategy than simpler strategies.

III. β Decomposition and Concentration Effect

The S&P 500 is commonly utilized as a benchmark for the market portfolio within the Capital Asset Pricing Model (CAPM). However, the disproportionate weighting of the MAG7 introduces significant bias into β estimations, leading to an overestimation of systematic risk for stocks highly correlated with MAG7 while underestimating it for others, potentially distorting risk assessments and asset pricing models. In 2023, the cohort generated a cumulative return of 75.7%, substantially outperforming the broader S&P 500, which returned only 24.2%. Notably, Nvidia alone experienced a surge of 239%, further amplifying its concentration risk.

To quantify impact of this explosive growth, we decompose the individual stock βs from the S&P 500(β_{500}) into contributions from the MAG7 (β_7) and the remaining 493 stocks (β_{493}). We also compare the explanatory and forecasting power of the traditional CAPM model with separated β model. To account for heteroscedasticity and time-varying correlation, we introduce the DCC-GARCH model and perform a stress test for unexpected shocks.

A. Data Description

This study uses the daily market capitalization and the daily closing prices of the S&P 500 index & Russell 3000 index and their constituent stocks, along with the ICB Industry classifications for all constituent stocks covering the period from January 2014 to December 2024.

B. β Decomposition

1) Methods for β decomposition: We construct a market capitalization weighted index by partitioning the market into two segments: The "MAG7" and the remaining 493 stocks. The daily returns of these segments are denoted by r_7 and r_{493}, respectively. r_{500} is the overall market return computed as the cap-weighted average of the 500 stocks. For calculating the contribution of MAG7 to individual stock β, we use the basic formula in CAPM:

$$\beta_i = \frac{\text{Cov}\,(r_i, r_{500})}{\text{Var}\,(r_{500})} = \frac{\text{Cov}\,(r_i, w_7 r_7 + w_{493} r_{493})}{\text{Var}\,(r_{500})}$$
$$= \frac{w_7\,\text{Cov}\,(r_i, r_7)}{\text{Var}\,(r_{500})} + \frac{w_{493}\,\text{Cov}\,(r_i, r_{493})}{\text{Var}\,(r_{500})}$$

We divide the β of individual stocks into two components, the first component representing the portion of β driven by MAG7, while the second component reflects the portion driven by the remaining 493 stocks. We used all 2,113 common stocks with available price data from 2014 and 2024. For each stock, we calculate its two β components within a 6-months rolling window and then take the simple average of all results.

2) Results for β decomposition: shows that the portion of individual stocks' β driven by MAG7 has trended upward in recent years. Assuming stocks are roughly homogeneous, over the long run, the average β contribution should approximate their weights but may deviate in the short-term due to market volatility, investor sentiment, and changes in individual stock correlations.

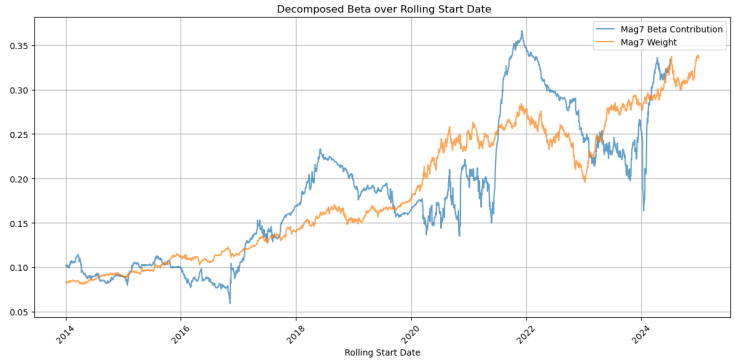


Fig. 1. MAG7 β Decomposition

Additionally, we have this across various sectors shows the MAG7 β for different industries over time, reflecting the trend in correlation between each industry and the MAG7. The Technology (TECH) sector has the highest β, that it is most closely related to the MAG7, especially spiking during the 2020 pandemic and the 2023 AI boom, while Utilities (UTIL) and Real Estate (REIT) have the lowest β. These results confirm that the MAG7's rise in market capitalization has led to a significant increase in the market's overall exposure to stocks in the technology sector.

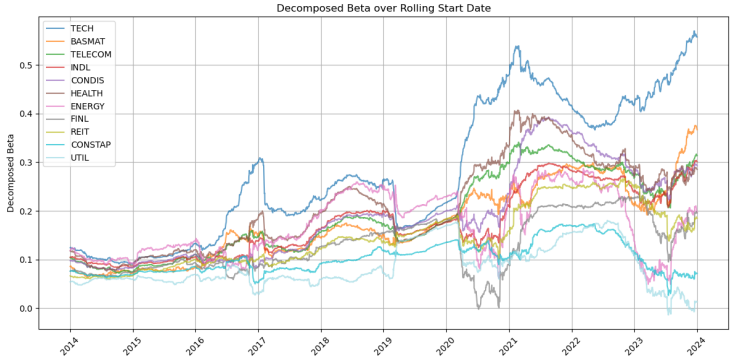


Fig. 2. Industry Based β Decomposition

We also calculated the β of stocks with the highest and lowest market capitalization, and the results (Figure 3) show that large-cap stocks have a higher β on MAG7 than small-cap stocks. As a result, the Long Small-Short Big strategy tends to have a negative β overall. It may underperform when the MAG7 is leading the market. This finding suggests that the traditional SMB (Small Minus Big) strategy can be improved through β adjustment or dynamic weight optimization to better adapt to changes in the market environment and improve robustness.

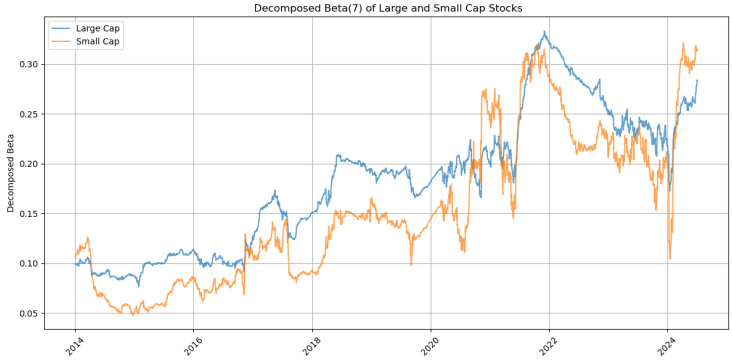


Fig. 3. Market Cap Based β Decomposition

C. Concentration Effect

1) Methods for Concentration Effect: Two competing models are considered. The first is the traditional Capital Asset Pricing Model (CAPM), which posits a single market factor for all stocks and is specified as follows:

$$\begin{aligned} r_i &= \beta_{500} r_{500} + \alpha_{500} + \epsilon_{500} \\ &= \beta_{500} w_7 r_7 + \beta_{500} w_{493} r_{493} + \alpha_{500} + \epsilon_{500}, \end{aligned} \tag{1}$$

where β_{500} is the market β, and α_{500} is the intercept.

The second model regresses individual stock returns on the returns of MAG7 and other stocks, isolating the effects of these two market segments:

$$r_i = \beta_7 r_7 + \beta_{493} r_{493} + \alpha_i + \epsilon_i. \tag{2}$$

$$H_0 : \ w_7\beta_{493} + (w_7 - 1)\beta_7 = 0, \ R = [w_7, 1 - w_7]', \ a = 0 \quad (3)$$

$$A = \left(R'\hat{\beta} - a\right)' \left(R'\hat{V}_{\hat{\beta}}R\right)^{-1} \left(R'\hat{\beta} - a\right) \xrightarrow{d} \chi^2_q$$

where w_7 and w_{493} denote the respective weights of MAG7 stocks and the remaining stocks in the index. Notably, the CAPM can be viewed as a restricted version of the second model under H_0 (3), since r_{500} can be rewritten as $(w_7 r_7 + (1 - w_7)r_{493})$. Rejection of this null hypothesis implies that separate βs for MAG7 and the remaining stocks are necessary rather than using a single aggregated market β.

We applied regression (2) and a Wald test to 2,113 individual stocks annually from 2014 to 2024, covering periods of low to high market concentration. To mitigate the effects of weight fluctuations over a year which may affect the test result, we use $w_7^* \in [min(w_7), max(w_7)]$ to denote the weight of the MAG7 stocks in the index that maximize the p-value of the Wald test each year.

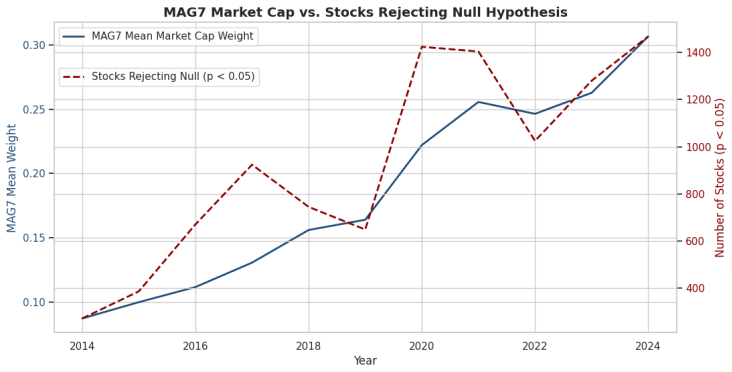


Fig. 4. Wald Test

2) Wald, F, LR, T Tests: Our empirical findings reveal an increasing number of cases where $p < 0.05$, suggesting that the imposed restriction of a single β becomes less applicable as market concentration increases over time. Similar patterns emerge from β_7 t-test results. We conclude that, as market concentration intensifies, distinguishing between β_7 and β_{493} yields a more accurate explanation of stock returns than relying solely on β_{500}.

Finally, we employ the F-tests and the log-likelihood ratio tests to compare the two competing models, assessing whether the additional predictors in the full model significantly improve its fit. Specifically, we calculate

$$F = \frac{(RSS_{\text{nested}} - RSS_{\text{full}})/p}{RSS_{\text{full}}/(n-k)}$$

and

$$\lambda = -2 \ln \left(\frac{L_{\text{nested}}}{L_{\text{full}}} \right).$$

We observe a strong correlation between the number of stocks rejecting the null hypothesis (H_0) and the increasing concentration of the MAG7 stocks, suggesting the necessity of a distinct β measure for mega-cap stocks in highly concentrated market environments. Under such conditions, the broader market is exposed to a new source of systematic risk stemming from mega-cap dominance, challenging the traditional CAPM assumption that systematic risk originates solely from the broader market. Accordingly, differentiating between β_7 (for mega-cap stocks) and β_{493} (for the remaining index constituents) offers a more accurate explanation of stock returns than relying solely on β_{500}.

D. Decorrelation and Ridge Regression

$$r_i = \beta_\epsilon \epsilon_7 + \beta_{493} r_{493} + \alpha_i + \epsilon_i, \quad (4)$$

$$r_7 = \beta_0 + \beta r_{493} + \epsilon_7. \quad (5)$$

We observe that the collinearity between r_7 and r_{493} can compromise the accuracy of the Wald test. To mitigate this, we further extract ϵ_7, the residuals from regressing r_7 on r_{493}, to ensure that ϵ_7 is orthogonal to r_{493}.

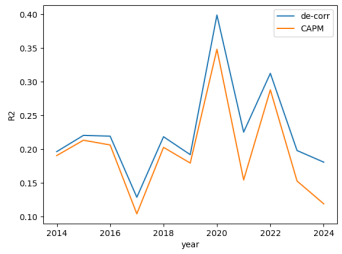


Fig. 5. R^2 of Decorr Model & CAPM

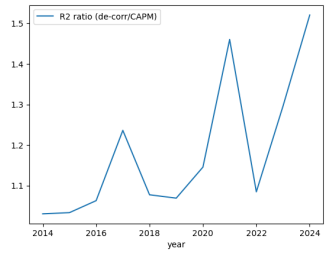


Fig. 6. Ratio of R^2 (de-corr/CAPM)

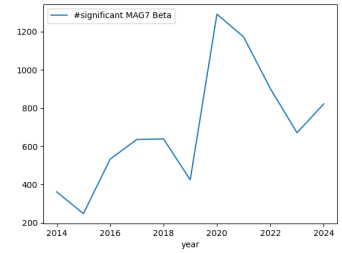


Fig. 7. NO. Significant β_ϵ (p < 0.05)

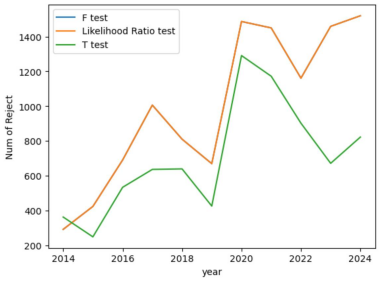


Fig. 8. F, LR, T tests

We apply regression (4) to our universe of 2,113 stocks. As demonstrated in Fig. 5. and Fig. 6., the separate model consistently achieves a higher R^2 than the CAPM model, with the R^2 ratio (separate model/CAPM) trending upward over time. In addition, Fig. 7. shows an increasing number of statistically significant β_ϵ estimates, indicating that the unique impact of MAG7 stocks on returns has grown over the years.

$$L(\beta) = \sum_{i=1}^{n} (y_i - X_i\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \quad (6)$$

For forecasting performance, we implement a 6-fold cross-validation to compute the Mean Squared Forecast Error (MSFE) for both models. To further alleviate the collinearity between r_7 and r_{493}, we extend our analysis by applying ridge regression as an alternative to model (2). We then compute

the ratio of CAPM MSFE to ridge MSFE, and compare the corresponding R^2 values. After 2019, this ratio (Fig. 9.) shows a modest increase, indicating that the forecasting error of the CAPM model worsened with rising market concentration. Moreover, as illustrated in Fig. 10., ridge exhibited a lower R^2 than CAPM before 2019 but consistently outperformed it thereafter, corroborating our earlier findings.

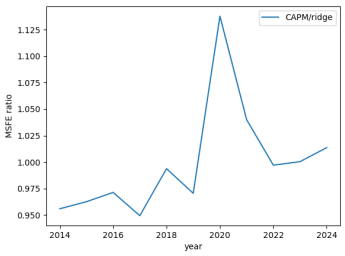


Fig. 9. Ratio of MSFE



Fig. 10. R^2 of CAPM and Ridge

E. DCC-GARCH and Shock Simulation

In the previous analysis, we assessed the ability of β to explain stock returns using linear regression. However, stock returns often exhibit autocorrelation and volatility clustering, which contradict the assumption in linear regression that explanatory variables maintain a static relationship with stock returns. Therefore, linear regression is likely insufficient to model how stock prices react to shocks, and we propose that time-series models may provide a more suitable framework.

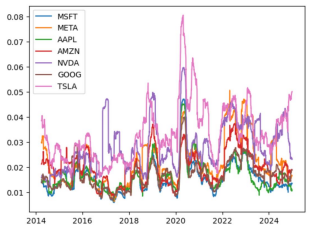


Fig. 11. Rolling Std. Dev Mag7

To simulate the impact of substantial price changes in MAG7 securities on other stocks, we model individual stock returns and their volatility dynamics. As shown in Fig. 11., the rolling standard deviation of MAG7 returns has fluctuated significantly over the past decade, indicating that stock returns exhibits heteroscedastic volatility. To capture volatility clustering, we consider a GARCH(1,1) process, ensuring that the model reflects the time-varying nature of return volatility. Additionally, we introduce an autoregressive (AR) term to prevent stock returns from being completely insensitive to introduced shocks, thereby enhancing the realism of the simulations by allowing for mean shifts following consecutive shocks.

$$r_t = \alpha_0 + \alpha_1 r_{t-1} + e_t, \quad e_t = \sigma_t \epsilon_t, \quad \epsilon_t \sim \text{i.i.d.}(0,1) \tag{7}$$

$$\sigma_t^2 = \alpha_2 + \alpha_3 e_{t-1}^2 + \alpha_4 \sigma_{t-1}^2 \tag{8}$$

where r_t represents the stock return at time t, and e_t is the error term with conditional variance σ_t^2. The parameter α_3 captures the impact of shocks on volatility, while α_4 measures the persistence of past volatility.

Stock returns exhibit dynamic relationships, and correlations between asset returns are not constant over time. Historically, the S&P 500 maintained a more balanced sector distribution, ensuring that even in the presence of market shocks, return correlations remained relatively stable. However, with the increasing dominance of technology-related stocks, historical correlation estimates are less reliable for analyzing evolving market dynamics. To address this limitation, we employ a DCC-GARCH model(Decker, Ferla, 2003), which captures time-varying correlations and allows for forward-looking stress tests.

$$Q_t = (1 - a - b)\overline{Q} + a\epsilon_{t-1}\epsilon_{t-1}' + bQ_{t-1} \tag{9}$$

$$R_t = \text{diag}(Q_t^{-1/2})\, Q_t\, \text{diag}(Q_t^{-1/2})$$

$$\overline{Q} = \frac{1}{T}\sum_{t=1}^{T} \epsilon_t \epsilon_t'$$

$$a > 0, \quad b > 0, \quad a + b < 1$$

where Q_t is the conditional covariance matrix of standardized residuals from the GARCH model and R_t is the correlation matrix derived from Q_t. The term $\overline{Q}$ represents the unconditional covariance of realized returns.

The DCC-GARCH model provides a dynamic framework for updating individual stock volatilities and their correlations. The recursive and closed-loop nature of this framework enables simulations of probable stock return paths in response to arbitrary market shocks. This methodology allows us to specify which stocks initiated a shock, the magnitude of the shock, and its duration.

The simulation process consists of the following steps:

1. Initialization: Set r_0 as the return vector of all stocks on December 31, 2024. Define ϵ_0 as the residual vector obtained from the fitted GARCH model on the same date. Set Q_0 and $\bar{Q}$ as the sample variance-covariance matrix for all stocks from January 2, 2014, to December 31, 2024. Define σ_0 as the sample standard deviation, obtained as the square root of the diagonal elements of the variance-covariance matrix.

2. Recursive Updates: Update the conditional covariance matrix using DCC-GARCH:

$$Q_{t+1} = (1 - a - b)\bar{Q} + a\epsilon_t\epsilon_t' + bQ_t$$

Normalize Q_{t+1} to derive the updated correlation matrix Σ. Update stock volatility dynamics using the GARCH equation:

$$\sigma_{t+1}^2 = \alpha_2 + \alpha_3 e_t^2 + \alpha_4 \sigma_t^2$$

Update stock expected returns using an AR(1) process:

$$\mu_{t+1} = \alpha_0 + \alpha_1 r_t$$

3. Shock Specification and Response Simulation: Define the shock event, specifying which stocks initiated the shock, the shock magnitude, and its duration. Assign the shock

adjusted returns: r_{shock}, and compute the corresponding residuals:

$$\epsilon_{\text{shock}} = \frac{r_{\text{shock}} - \mu_{\text{shock}}}{\sigma_{\text{shock}}}$$

Assuming that the stock returns follow a multivariate normal distribution, determine the conditional distribution of the unaffected stocks given the shocked stocks:

$$\epsilon_{\text{sim}} | \epsilon_{\text{shock}} \sim$$

$$\mathcal{N}\left(\Sigma_{12}\Sigma_{22}^{-1}(\epsilon_{\text{shock}}), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right)$$

Sample from this distribution to obtain simulated residuals ϵ_{sim}, and compute final simulated returns:

$$r_{\text{sim}} = \mu_{\text{sim}} + \sigma_{\text{sim}}\epsilon_{\text{sim}}$$

This framework provides a robust mechanism for simulating market reactions to idiosyncratic and systemic shocks, offering insights into risk dynamics, correlation shifts, and volatility spillover effects. We initiated a 2 standard deviation shock from Nvidia and simulated how other stocks respond as illustrated in Fig. 12. From the simulated volatility, we observe an increasing trend during the first week after the shock, with the impact gradually diminishing thereafter. This indicates a volatility spillover effect during the initial week. The shock initially triggers a sudden rise in the residual term of the affected stocks, causing the next day's volatility to increase due to GARCH updating. This volatility then spills over to other stocks through the conditional distribution of the residual term, leading to a rise in both the current day's return and the next day's volatility, again due to GARCH updating. The increase in the residual term also raises the correlation between stocks through DCC updating, thereby amplifying the volatility spillover effect and creating a positive feedback loop. Conversely, the impact of the volatility does not escalate indefinitely but rather subsides after the first week. A negative shock decreases the conditional mean of return via the AR term while increasing conditional volatility. This results in a reduction in the residual term the following day, thereby mitigating the volatility spillover effect. This demonstrates how market participants adjust their beliefs based on recent events, making subsequent shocks less impactful.

This phenomenon illustrates that during prosperous times, individual stock volatility and inter-stock correlations are low. However, during adverse events, both individual stock volatility and correlations rise significantly, exacerbating the situation. It takes time for the market to fully incorporate the shock, and the impact of the shock gradually diminishes.

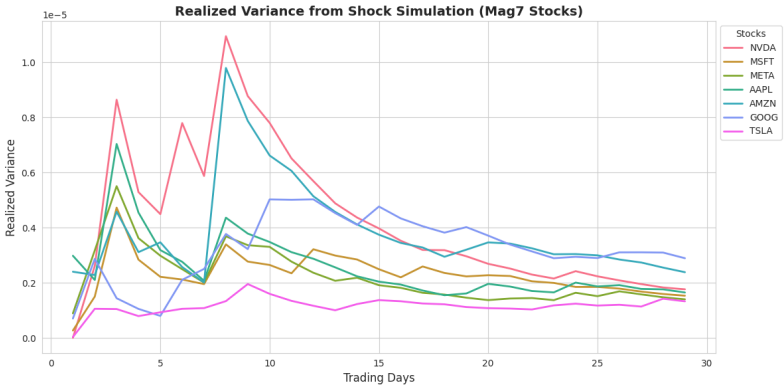


Fig. 12. Realized Variance from Shock Simulation

IV. Options Pricing and Correlation Risk

This section further examines the impact of market concentration through the lens of the options market, focusing on its implications for risk pricing and investor sentiment. In particular, we examine empirical evidence of an additional risk premium associated with the heightened perceived probability of a market downturn due to increased concentration in the S&P 500. Building on this, we develop a framework to estimate the impact of increased volatility in underlying securities on S&P 500 index options.

A. Data Description

This study utilizes daily market capitalization and closing prices of the S&P 500 index and its constituent stocks, along with end-of-day bid and ask prices for call and put options on S&P 500 constituents from January 2014 to August 2023. Additionally, we retrieve the daily treasury par yield curve from the U.S. Department of the Treasury.

B. Methodology

1) Single-Stock VIX (SSVIX): We propose a robust methodology, similar to that of the CBOE S&P 500 Volatility Index (VIX) (CBOE, 2025), to derive a constant 30-day SSVIX that captures the expected risk-neutral volatility of individual stocks. To model the risk-free rate across different maturities, we apply cubic spline interpolation to the Treasury yields dataset. Subsequently, the methodology categorizes equity options data into distinct groups based on their Day-To-Expiry (DTE). To mitigate numerical inaccuracies arising from limited option availability, we first categorize options into four groups: two with a DTE of less than 30 days and two with a DTE of 30 days or more. For each maturity, the contributions to the overall variance are estimated using a weighted mean of the option premiums $\frac{1}{K^2}$, where the weights represent the difference between the strike price and the risk-free discount factor. These DTE-specific variance values are then linearly interpolated to derive a SSVIX value for each trading day. We confirm that the replicated VIX values closely align with the industry standard CBOE Equity VIX series (e.g., VXAPL), with an average R-square of 0.99.

2) Implied Correlation: We propose an enhanced methodology based on the CBOE Implied Correlation Methodology (CBOE, 2022), where we replace the conventional at-the-money implied volatility with the SSVIX of the top 50 index constituents. Under this framework, the 30-day implied correlation is computed using the following expression:

$$\bar{\rho} = \frac{\sigma^2_{\text{S\&P 500}} - \sum_{i=0}^{49} w_i^2 \sigma_i^2}{2 \sum_{i=0}^{49} \sum_{j>i} w_i w_j \sigma_i \sigma_j}, \tag{10}$$

where $\bar{\rho}$ denotes the implied correlation, σ_i represents the SSVIX of the ith constituent stock, and w_i is the rescaled weight of the ith stock, assuming the index consists of the top 50 stocks

3) CCC EGARCH: We employ a one-factor EGARCH model, as proposed by Engle and Figlewski (2014), to model the dynamics of SSVIX and correlation for the top 50 stocks. For each stock i, the model for the evolution of its SSVIX is given by

$$\log\left(\frac{v_{i,t}}{v_{i,t-1}}\right) = \theta_{1,i} + \theta_{2,i} \log\left(\frac{\text{VIX}_t}{\text{VIX}_{t-1}}\right)$$

$$+\theta_{3,i} \log(v_{i,t-1}) + \theta_{4,i} \log(\text{VIX}_{t-1}) + \sigma_{i,t}\, \epsilon_{i,t}, \tag{11}$$

$$\log\left(\sigma^2_{i,t}\right) = \phi_{1,i} + \phi_{2,i} \left|\epsilon_{i,t-1}\right| + \phi_{3,i}\, \epsilon_{i,t-1} + \phi_{4,i} \log\left(\sigma^2_{i,t-1}\right), \tag{12}$$

The corresponding EGARCH specification for the VIX is

$$\log\left(\frac{\text{VIX}_t}{\text{VIX}_{t-1}}\right) = \theta_{1,\text{VIX}} + \theta_{2,\text{VIX}} \log\left(\text{VIX}_{t-1}\right) + \sigma_{\text{VIX},t}\, \epsilon_{\text{VIX},t}, \tag{13}$$

$$\log\left(\sigma^2_{\text{VIX},t}\right) = \phi_{1,\text{VIX}} + \phi_{2,\text{VIX}} \left|\epsilon_{\text{VIX},t-1}\right| +$$

$$\phi_{3,\text{VIX}}\, \epsilon_{\text{VIX},t-1} + \phi_{4,\text{VIX}} \log\left(\sigma^2_{\text{VIX},t-1}\right). \tag{14}$$

where $v_{i,t}$ denotes the SSVIX of stock i at time t and $\epsilon_{i,t}$ follows a t-distribution with ν_i degrees of freedom. This model is selected for its ability to incorporate broad market information and capture volatility's asymmetric response to shocks. Furthermore, using t-distributed innovations allows the model to capture the fat-tailed nature of the volatility process. The EGARCH model is estimated using Maximum Likelihood Estimation (MLE) based on the volatility measures derived above. Furthermore, a multivariate t-distributed constant conditional correlation (CCC) model is calibrated on the standardized residuals of the EGARCH estimation. Given the high dimensionality of the parameter space, we employ the Expectation-Maximization algorithm to efficiently obtain the parameter estimates.

C. Empirical Implications

In this section, we evaluate the impact of market concentration on option pricing.

1) VIX, Realized Volatility (RV), Implied Correlation (IC), and Realized Correlation (RC): As shown in Figure 13, VIX levels do not vary significantly with increasing market concentration over the 2014–2024 period. Similarly, the spread between IC and RC remains predominantly positive, although noisier than the volatility spread. It is a common perception that increase in market concentration leads to higher systematic risk and potential market crash, thus expensive options. However, actual market data tells a different story. One possible explanation for this counterintuitive outcome arises is that the market assigns greater weight to high-quality blue-chip stocks with sustainable business models and robust profitability, thereby stabilizing overall market risk.

In the post-COVID era, increased market concentration has led to a modest increase in the mean VIX level, while the IC remains relatively flat. As MAG7 stocks, which typically exhibit higher implied volatility, dominates the market portfolio, index options inherit the characteristics of these stocks, results in higher VIX on average.

The volatility and correlation spreads remain largely stable under varying concentration levels. As depicted in the charts, both spreads are consistently positive under normal conditions, reflecting an insurance premium for non-hedgeable volatility and correlation risk. To further support this observation, we perform an OLS regression of these spreads on changes in the total weight of MAG7 stocks and a one-period lag of the spread. The regression results (Table I) indicate that changes in weight are not statistically significant, suggesting that concentration risk is not priced in despite the growing weight of the MAG7.

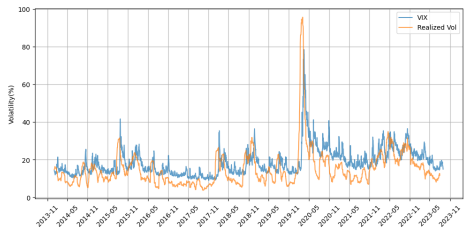


Fig. 13. VIX and Realized Volatility (2014-2023)

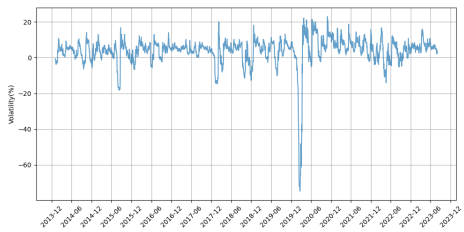


Fig. 14. Volatility Spread

TABLE I
OLS Regression Results

	(1) Volatility Spread	(2) Correlation Spread
Weight	-0.0044 (0.007)	2.2908 (2.035)
Difference (Lagged)	0.0096*** (5.39e-05)	0.9268*** (0.008)
Constant	-0.0006	0.1083
R-Squared	0.931	0.859

Standard errors in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

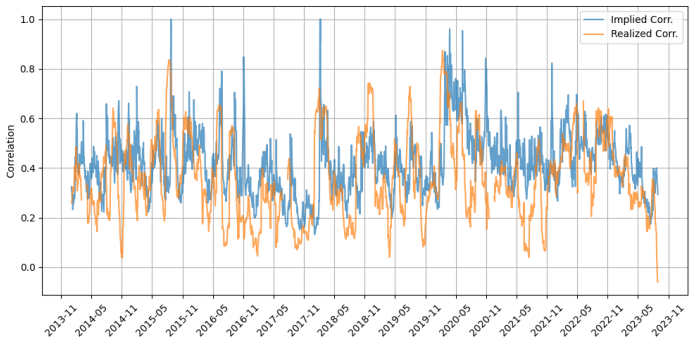


Fig. 15. Implied Correlation and Realized Correlation (2014-2023)

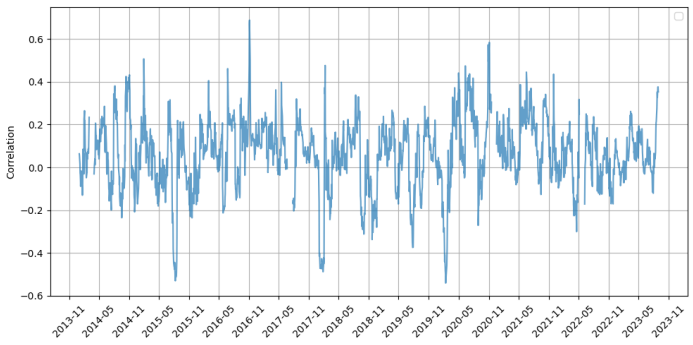


Fig. 16. Correlation Spread

D. Analytical Framework

Our results in the previous section demonstrate that options have not become significantly more expensive despite the growing presence of MAG7 within the index. These stocks are not immune to adverse news. For example, if breaking news were to negatively affect MAG7, index implied volatility would likely be high due to their significant risk concentration and large index weights, which in theory should increase option premium. To investigate this, we employ our fitted CCC-EGARCH model to construct a double ρ SSVIX-based estimator that captures both the correlation between (log-)implied volatilities and that between underlying returns. The analytical framework is outlined as follows:

1) Assume that n assets experience shocks. For these assets, determine the pth percentile of the historical standardized residuals of log SSVIX and return innovations.
2) Derive the conditional distribution of the standardized residuals for non-shocked assets, given the standardized residuals of the n shocked assets and the underlying correlation structure. For example, following Ding (2016), the conditional distribution of a t-distributed variable is given by:

$$u_{1|2} = \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2,$$

$$\Sigma_{11|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21},$$

$$d_2 = \mathbf{X}_2^T\Sigma_{22}^{-1}\mathbf{X}_2,$$

$$\mathbf{X}_{1|\mathbf{X}_2} \sim t\left(u_{1|2}, \frac{\nu + d_2}{\nu + p_2}\Sigma_{11|2}, \nu + p_2\right),$$

where $\mathbf{X}1$ and $\mathbf{X}_2$ denote the vectors of unknown and given observations, respectively; Σ_{ij} represents the (i, j) partition of the scale matrix; and p_2 is the length of the given observation.
3) Simulate m realizations of the standardized residuals for returns and volatilities of the non-shocked assets.
4) Compute the SSVIX for all 50 constituents using the simulated standardized residuals and the fitted EGARCH model.
5) Update the mean correlation estimate $\bar{\rho}$ based on the simulated return innovation and the fitted DCC-GARCH from Section III-E.
6) For each simulation, calculate the VIX using the SSVIX of all constituents, their weights, and the updated correlation:

$$\sigma^2_{\text{S\&P500}} = \sum_{i=0}^{49} w_i^2\sigma_i^2 + 2\sum_{i=0}^{49}\sum_{j>i} w_i w_j \sigma_i \sigma_j \bar{\rho}.$$

For this analysis, we consider several hypothetical scenarios to assess the expected change in VIX levels. We extract options parameters from the market data at 31 Aug 2023. The option parameters are set as follows: risk-free rate $r = 5.5\%$, dividend yield $y = 0\%$, spot price $S = 4508$, ATM implied volatility $\sigma_{\text{imp ATM}} = 11.5\%$, and 5% OTM implied volatility $\sigma_{\text{imp 5\%OTM}} = 16.1\%$. We estimate that during a crisis the index drops by 2.5% with the return innovations and fitted DCC-GARCH model, and the implied volatility curve shifts in parallel, which aligns with our observations. This implies $\Delta\text{VIX} = \Delta\sigma_{\text{imp}}$. The results are summarized in Table II.

TABLE II
SHOCK ANALYSIS RESULTS AT THE 99TH PERCENTILE

Shock Asset	n	$\Delta\sigma_{\text{imp}}$	% Change in Put Price	
			ATM	5% OTM
Any	1	4.21	168%	324%
Any	3	5.88	184%	387%
Any	5	6.75	192%	421%
Any	7	7.45	198%	448%
MAG7	1	4.44	170%	332%
MAG7	3	5.96	184%	390%
MAG7	5	6.73	148%	492%
MAG7	7	7.13	151%	508%

Table II clearly shows that significant shocks to one or more stocks trigger volatility spillovers across the market, thereby increasing put option prices. When a shock is applied to a single asset, diversification mitigates the impact on index options. However, as the number of shocked assets increases, idiosyncratic risk converts into systematic risk, leading to a pronounced rise in the index's implied volatility. Notably, the impact of shocks to MAG7 stocks is significant even when only one MAG7 stock is affected. This is due to the high correlation between the implied volatility of the shocked asset and the implied volatilities of other assets. This indicates that in a more homogeneous market, a shock to a single asset can readily propagate across the broader system, elevating the implied volatility of index options.

In such an environment, where concentration risk heightens implied volatility interdependencies across assets, options Vega exposure needs to be considered by market participants before taking any positions. Market concentration has created a highly correlated market, where shocks to one stock are likely to trigger similar reactions in others, increasing the likelihood of a simultaneous increase in implied volatility. Although options market makers manage the vega profile of their options book through taking offsetting positions, are still exposed to residual vega risk. They must reassess their hedging strategy by better incorporate the effect of increased implied volatility correlation, to protect themselves against market shocks.

V. Trading Strategy

In this section, we examine the impact of market concentration on long-short trading strategies by analyzing the performance of Small Minus Big (SMB) portfolios under current market conditions relative to their historical performance.

A. Background

Empirical evidence has long contested the Capital Asset Pricing Model (CAPM), which asserts that, in a well-diversified portfolio, market risk is the sole determinant of the risk-return tradeoff. Therefore, Fama and French introduced their three-factor model adding size and value factors to the CAPM mode to better explain cross-sectional variations in returns. However, Recent studies suggest that the explanatory power of the SMB factor has diminished. In hindsight there could be multiple reasons behind this: the rise of passive investing and strong revenue streams from mega-cap stocks have created the impression that the SMB factor is no longer effective. The SMB factor, capturing the return differential between small and large-cap stocks, is intrinsically linked to the HML factor, which reflects the value–growth spectrum. However, MAG7 stocks propelled by robust future earnings expectations from innovations in AI and autonomous driving exhibit a decoupling of the expected size–value relationship. This divergence casts doubt on the explanatory power of the SMB factor for capturing size-related return premia. At the same time, heightened investor risk aversion amid the current stagflationary environment has further reduced the small-cap premium, reinforcing this perception. Therefore, the short leg of the value-weighted SMB portfolio is implicitly betting against the large industries and mega-cap stocks like the MAG7, effectively shorting the β to the S&P 500 index, as the mega-cap stocks have become highly correlated with the market due to increased concentration.

Beyond the standard decomposition of market β, our options pricing analysis indicates that systematic risk stemming from the MAG7 deserves to be explicitly incorporated within the SMB framework. While concentration risk is not immediately evident given the stability of option premiums, the prevailing market conditions characterized by low volatility and abundant liquidity, create an environment conducive to increased concentration. In the absence of a significant market downturn and with a subdued VIX, the detrimental effects of concentration are yet to manifest. However, a negative shock to the MAG7 could propagate on a scale comparable to broader systematic risk. Consequently, neutralizing β_7 is arguably crucial for effectively isolating and capturing returns attributable to the SMB factor. Given the substantial weight of the MAG7 within the S&P500, this approach would also mitigate overall market β exposure.

Given the high concentration of the MAG7 stocks, these securities—and those highly correlated with them—can act as central triggers for a market-wide fire sale, as evidenced by our options market analyses and the systematic risk decomposition via β_{500}. This dynamic could precipitate rapid and severe price declines. To mitigate such systemic risk, it would be imperative to maintain exposure to stocks that are neutral relative to the MAG7. In light of these structural shifts, traditional SMB portfolios require re-evaluation and adjustment to incorporate strategies that effectively manage and mitigate concentration risk.

B. Hypothesis

We hypothesize that the current market and sector concentration are detrimental to the performance of conventional SMB portfolios.

C. Data Description

This study employs daily market capitalization data and closing prices of the S&P 500 index and its constituent stocks, covering the period from March 1957 to December 2024. We use the ICB Industry classification from WRDS.

D. Methodology

1) Market Concentration Indices: We adopt market concentration measures inspired by the labor market inequality literature (e.g., Pandey et al.) and apply them to equity markets. Specifically, we use the Hirschman-Herfindahl Index (HHI) as an absolute measure of concentration and the Gini Index as a relative measure of inequality.
Hirschman-Herfindahl Index (HHI) -

$$HHI = \sum_{i=1}^{n} s_i^2$$

where s_i: The market share of the ith stock, typically expressed as a fraction of the total market capitalization. Specifically,

$$s_i = \frac{x_i}{\sum_{j=1}^{n} x_j},$$

and x_i is the market capitalization of the ith stock.
Gini Index -

$$G = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2n^2 \bar{x}}$$

$\bar{x}$ The average market capitalization, given by:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

We then examine the correlation between six-month returns of the SMB portfolio and the change in the Gini Index and HHI over the corresponding period, while also assessing the persistence of these concentration measures.

2) Diversification of SMB: In periods of high market concentration, mega-cap stocks tend to dominate the short leg of traditional SMB portfolios, thus increasing the volatility of the general portfolio and their exposure to a small subset of stocks. To address this issue, we propose a three-step framework. First, we make the portfolio industry-neutral by selecting the top and bottom stocks within each industry based on market capitalization. Second, we transition from a value-weighted portfolio to an equal weighted portfolio. Third, we neutralize exposure to the MAG7 portfolio by adjusting the weights of the long and short positions based on β_7 estimates calculated for MAG7 stocks.

E. Results

We observe a strong negative correlation -0.87 between monthly SMB portfolio returns and changes in the Gini Index (a proxy for market concentration) and a negative correlation of -0.52 with HHI (the absolute measure of concentration). This relationship is intuitive, as periods of large-cap stock out-performance relative to small-cap stocks correspond to rising market concentration. Furthermore, market concentration changes exhibits significant persistence, with a six-month autocorrelation of 0.87. Leveraging this persistence, we classify market regimes based on long term trends in concentration (See Fig. 17.).

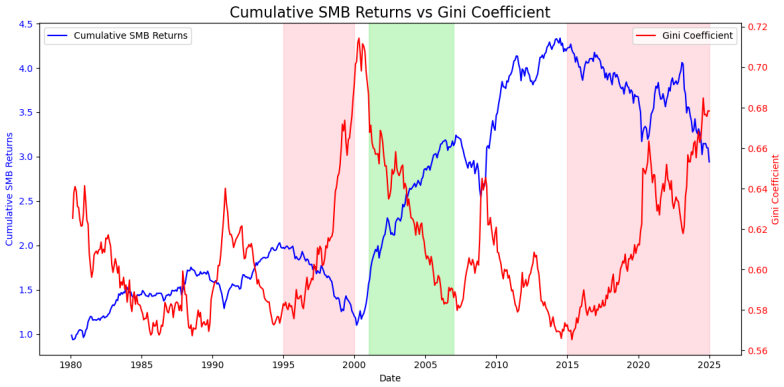


Fig. 17. Increasing/Decreasing Market Concentration Regimes based on Gini

From 1995 to 2000, during the dot-com bubble, market concentration increased significantly, coinciding with a -50% cumulative return (-11% annualized) for the SMB portfolio. Conversely, from 2001 to 2007, a period of declining concentration, the SMB portfolio delivered a cumulative return of 300% (20% annualized). These findings suggest that the SMB portfolio generates superior returns in regimes of declining concentration. The persistence of concentration trends further implies that such regimes can persist for extended periods, offering exploitable opportunities.

A potential enhancement to the SMB strategy involves dynamically inverting its exposure in regimes of increasing concentration. However, the efficacy of this approach is contingent on accurately timing regime shifts, which presents challenges due to significant transaction costs associated with completely inverting our portfolio and the unresolved high exposure to mega-cap stocks such as the MAG7 stocks.

Instead, we propose an alternative approach to mitigate regime dependency by adjusting the portfolio's long and short legs dynamically. Specifically, we construct a modified SMB portfolio that neutralizes its β exposure to Mag7 stocks and reduces industry exposure through selecting the same number of stocks from each industry.

Feghali and Villalon (2015) emphasized the role of industry-neutral portfolio construction in mitigating concentration risk, particularly in markets increasingly dominated by a small subset of stocks such as the MAG7, which are concentrated within a few sectors. Inadvertently, market participants allocating capital without industry constraints may be disproportionately exposed to these sectors, effectively placing implicit bets on their continued outperformance. This heightened exposure increases systematic risk, as portfolio returns become more dependent on sector-specific factors rather than idiosyncratic stock selection.

By employing an industry-neutral framework, investors can systematically control for sectoral imbalances, ensuring that excess returns are derived from stock-specific alpha rather than the aggregate performance of highly concentrated industries. This approach not only reduces immediate exposure to dominant sectors but also future-proofs portfolios against potential shifts in market leadership. As market concentration intensifies, industry-neutral strategies help maintain a balanced risk profile, mitigating the adverse effects of sector-driven crowding and ensuring a more stable and diversified return stream over time.

Additionally, we use an equal-weighted portfolio. The study by Edwards, Lazzara, Preston, and Pestalozzi (2018) examines the effects of equal-weighted portfolio construction and its implications for diversification, risk management, and long-term returns. Unlike market-capitalization-weighted indices, which allocate more weight to larger companies and can lead to sectoral concentration, equal-weighted indices assign an even allocation across all constituents, reducing the influence of any single stock or industry. This structure helps to limit industry-specific biases and promotes a more balanced exposure across sectors.

In addition, the rebalancing mechanism in equal-weight indices systematically adjusts portfolio allocations, ensuring that capital is redistributed from stocks that have increased in value to those that have declined. This process preserves the stock selection SMB signal of our trading strategy.

Our empirical analysis reveals that the modified portfolio construction significantly outperforms the traditional SMB strategy. Specifically, the modified strategy achieves Sharpe ratios exceeding 1.1 (Fig. 18.) in both pre-COVID and post-COVID periods in comparison with the traditional SMB portfolio with a Sharpe Ratio of near zero. Moreover,

by reducing exposure to the MAG7 stocks, the portfolio demonstrates enhanced resilience to the MAG7 systematic shocks, as evidenced by its immunity to the Tesla shock observed in 2024 (Fig. 19.).

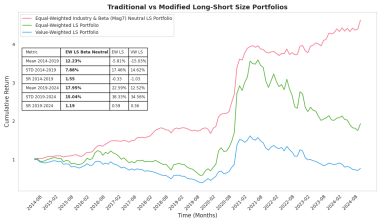


Fig. 18. Cumulative Returns of Traditional SMB vs Modified SMB

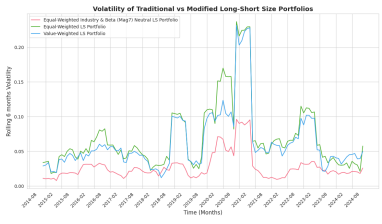


Fig. 19. Rolling volatility of Traditional SMB vs Modified SMB

Our modified strategy consistently achieves a significant reduction in volatility across both periods. We observe the most significant improvement in maximum drawdown (Fig. 20.). Under the traditional SMB strategy, the maximum drawdown reached -60%, whereas our modified SMB strategy reduced it to -8%.

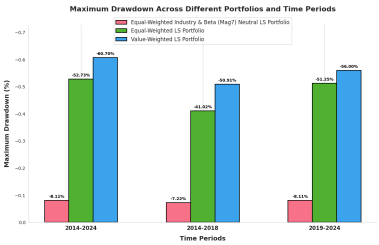


Fig. 20. Maximum Drawdown of Traditional SMB vs Modified SMB

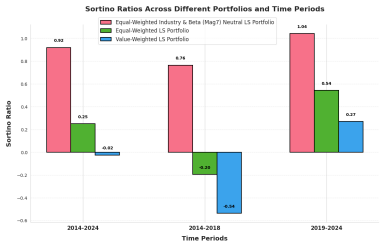


Fig. 21. Maximum Drawdown of Traditional SMB vs Modified SMB

As illustrated in Fig. 21. the Sortino Ratio, which focuses on downside risk to measure the risk-return tradeoff, is 0.92 for the modified SMB strategy compared to -0.02 for the traditional SMB strategy.

VI. Conclusion

Our analysis demonstrates a clear need for a paradigm shift in market risk assessment. β decomposition and comparison between CAPM and seperated β model shows that MAG7 constitute a seperate risk factor. Simulation of DCC-GARCH implies potential volatility spillover effect in the stock market within the first week after shock. We provide empirical evidence of correlation risk in the options market and argue for broad adjustments to traditional long-short strategies to address both correlation risk and the systematic risk associated with the MAG7. Thus, this DCC-GARCH result can improve the risk management strategy for trading. Furthermore, market concentration heightens implied volatility interdependencies. In this environment, managing options vega exposure while accounting for implied volatility correlation is crucial to protect against potential systematic shock.

Future research should focus on quantifying the global influence of the MAG7 and its potential spillover effects on international markets. Advanced machine learning approaches, such as transformer-based architectures, could be employed to capture the nonlinear dependencies between MAG7 stocks and macroeconomic indicators. Additionally, behavioral studies may shed light on the role of retail investor herding in exacerbating market concentration, while regulatory research might explore whether antitrust measures or sector-specific position limits could help mitigate systemic risks.

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